

PAPER

Theory of polarization-averaged core-level molecular-frame photoelectron angular distributions: I. A full-potential method and its application to dissociating carbon monoxide dication

To cite this article: F Ota *et al* 2021 *J. Phys. B: At. Mol. Opt. Phys.* **54** 024003

View the [article online](#) for updates and enhancements.

You may also like

- [C 1s photoelectron angular distributions from fixed-in-space CO molecules in the high-energy continuum 50 eV](#)
Jun-ichi Adachi, Misato Kazama, Takahiro Teramoto *et al.*
- [Molecular frame photoelectron angular distributions for core ionization of ethane, carbon tetrafluoride and 1,1-difluoroethylene](#)
A Menssen, C S Trevisan, M S Schöffler *et al.*
- [Asymmetry in the molecular-frame photoelectron angular distribution for oxygen 1s photoemission from CO₂](#)
R R Lucchese, H Fukuzawa, X-J Liu *et al.*

Theory of polarization-averaged core-level molecular-frame photoelectron angular distributions: I. A full-potential method and its application to dissociating carbon monoxide dication

F Ota¹, K Yamazaki^{2,6}, D Sébilleau³, K Ueda⁴ and K Hatada^{5,*} 

¹ Graduate School of Science and Engineering for Education, University of Toyama, Gofuku 3190, Toyama 930-8555, Japan

² Institute for Materials Research, Tohoku University, 2-1-1 Katahira, Aoba-ku, Sendai 980-8577, Japan

³ Département Matériaux Nanosciences, Institut de Physique de Rennes, UMR UR1-CNRS 6251, Université de Rennes, F-35000 Rennes, France

⁴ Institute of Multidisciplinary Research for Advanced Materials, Tohoku University, Katahira 2-1-1, Aoba-ku, Sendai 980-8577, Japan

⁵ Faculty of Science, Academic Assembly, University of Toyama, Gofuku 3190, Toyama 930-8555, Japan

E-mail: hatada@sci.u-toyama.ac.jp

Received 31 August 2020, revised 17 November 2020

Accepted for publication 2 December 2020

Published 01 February 2021



Abstract

We present a theoretical study of the polarization-averaged molecular-frame photoelectron angular distributions (PA-MFPADs) emitted from the $1s$ orbital of oxygen atoms of dissociating dicationic carbon monoxide CO^{2+} . Due to the polarization average, the contribution of the direct wave of the photoelectron, which represents the largest contribution to the MFPADs, is removed, so that the PA-MFPADs clearly show the details of the scattering image of the photoelectron. As a result, it is necessary to employ an accurate theory for the theoretical analysis of the continuum state. In this study, we apply a full-potential multiple scattering theory, where the space is partitioned into Voronoi polyhedra and truncated spheres, to take into account the electron charge density outside the physical atomic spheres. We do not use the spherical harmonic expansion of the cell shape functions to avoid divergence problems. The potentials in the scattering cells are computed using the multiconfigurational second-order perturbation theory restricted active space method to take into account the influence of the core hole in the electron charge density in the final state, so that a realistic relaxation can be achieved. We show that the full-potential treatment plays an important role in the PA-MFPADs at a photoelectron kinetic energy of 100 eV. In contrast, the PA-MFPADs are not sensitive to any type of major excited states in the Auger final state. We also study the dynamics of the CO^{2+} dissociation. We find that the PA-MFPADs dramatically change their shape as a function of the C–O bond length.

Keywords: PA-MFPADs, multiple scattering theory, full potential, FEL
(Some figures may appear in colour only in the online journal)

* Author to whom any correspondence should be addressed.

⁶ Present address: Attosecond Science Research Team, Extreme Photonics Research Group, RIKEN Center for Advanced Photonics, RIKEN, 2-1 Hirosawa, Wako, Saitama, 351-0198, Japan.

1. Introduction

Core-level photoelectron diffraction (PED) is a well-known technique used to probe surface structures [1]. To apply PED to gas-phase molecules, the molecule must be fixed in space. Experimentally, this can be realized by angle-resolved coincidence measurements between core-level photoelectrons and fragment ions [2]. With the advent of position sensitive detectors [3] and the COLTRIMS-reaction microscope [4], the measure of momentum correlations between core-level photoelectrons and fragment ions to extract the photoelectron angular distributions in the molecular frame has become a standard approach to the study of molecular photoionization, see, e.g. the first paper on molecular-frame photoelectron angular distributions (MFPADs) measured by this technique [5].

From a theoretical point of view, it is rather straightforward to calculate MFPADs with the codes available for molecular photoionization studies. For the theoretical tools for molecular photoionization studies in general, see, e.g. [6, 7]. One of the widely used approaches used to calculate MFPADs is to use relaxed-core Hartree–Fock (HF) continuum orbitals in the final-state wavefunction [8, 9]. The continuum orbitals may be obtained by iteratively solving the Lippmann–Schwinger equation associated with the one electron Schrödinger equation with a potential produced by the transition-state orbital [10, 11]. To take account of electron correlations, Semenov *et al* [12, 13] employed random-phase approximations, while Lucchese and collaborators employed multichannel Schwinger configuration interaction methods [14, 15]. Density functional theory (DFT) and time-dependent DFT (TDDFT) [16] have also been used to simulate MFPADs [17, 18].

At higher photoelectron energies, e.g. ~ 100 eV and above, the multiple scattering method within the muffin-tin approximation, which is a standard tool for PED from solid surfaces [19], has also been employed for calculations of the MFPADs of the gas phase molecules [20]. The multiple scattering method within the muffin-tin approximation is a powerful tool due to the good convergence of the angular momentum multi-center expansion of the local numerical basis. Therefore, this method allows one to treat relatively large polyatomic molecules. We find, however, that it gives a non-negligible error in the MFPADs, especially in the forward-scattering directions. This is because the muffin-tin approximation neglects the covalency and weak potential features out of the sphere, which contributes to the small-angle scattering. In the present study, to improve the multiple scattering method beyond the muffin-tin approximation, we introduce a recently developed full-potential approach, which fills up the space with Voronoi polyhedra and truncated spherical cells. This allows one to take into account the electron charge density outside the physical atomic cells of the target molecule. Note that the cell shape function is not expanded into spherical harmonics to avoid convergence problems. This approach has been successfully applied to x-ray absorption near edge structure (XANES), and has led to substantial improvements

in low-dimensional systems which have a strong anisotropy [21–23]. These improvements in muffin-tin calculations were observed in particular in the low-energy region (about 50 eV from the edge). In the higher-energy region, $\gtrsim 100$ eV, there was no noticeable difference observed between the muffin-tin and the full-potential XANES calculations.

It should be noted that the MFPADs depend on both the molecular axis and the direction of polarization of the ionizing radiation. Recently, it was observed that the molecular structural information is captured in the polarization-averaged MFPADs (PA-MFPADs). Due to the polarization-averaging, the contribution of the direct wave of the photoelectron, which is the largest contribution to the MFPADs, is removed so that the PA-MFPADs clearly show the scattering image of the photoelectron. Williams *et al* [24] were the first to demonstrate that PA-MFPADs capture the directions of the bonds in polyatomic molecules (CH_4). Stimulated by this experimental work, Plésiat *et al* [17] calculated the PA-MFPADs for many different molecules in a wide range of photoelectron energies, using DFT. Fukuzawa *et al* [25], on the other hand, demonstrated that one can extract the bond length of the symmetric linear molecule (CO_2) from the PA-MFPADs.

The advent of X-ray Free Electron Laser (XFEL) [26] has stimulated a new surge of interest in the study of MFPADs since time-resolved MFPADs or PA-MFPADs using XFEL pulses as ionizing pulses may open a new pathway to make molecular movies. Using a plane wave approximation at an electron kinetic energy of 1 keV, Krasniqi *et al* [27] theoretically demonstrated that the light-induced structural changes in polyatomic molecules can be extracted from the MFPADs. Kazama *et al* [20], on the other hand, used the multiple scattering method within the muffin-tin approximation to demonstrate that it is possible to extract the molecular structure from experimental MFPADs recorded at electron kinetic energies of about 100 eV, thereby opening the way to time-resolved MFPAD measurements with Free Electron Laser (FEL). Very recently, at the soft x-ray beam line of the European XFEL [28], the first high-repetition-rate XFEL, Kastirke *et al* [29] performed the first successful PA-MFPAD measurements at electron kinetic energies of ~ 100 eV, using diatomic molecules O_2 as a sample, thereby demonstrating the feasibility of making molecular movies by using a combination of a COLTRIMS-reaction microscope and a high repetition rate XFEL.

In the present work, we consider the process where the first XFEL pulse produces CO^{2+} via the Auger decay that follows the core ionization of CO and then, the second XFEL pulse kicks out an oxygen $1s$ electron from CO^{2+} . In sections 2 and 3, we derive the expressions of the PA-MFPADs for core-level photoemission within the full-potential multiple scattering theory. The results and their discussion are given in section 4. We demonstrate then that the PA-MFPADs are not sensitive to the electronic structure, which may differ depending on the Auger final states. Employing a particular electronic configuration of the dicationic state, we show how the PA-MFPADs evolve as a function of the internuclear distance. Finally, we compare the

present results to those obtained within the more commonly used muffin-tin approximation. This highlights the importance of a proper description of the scattering potentials.

2. Theory

2.1. Full-potential multiple scattering theory

We start from the identity obtained from the Green's theorem for the free electron Green's function $G_0^+(\mathbf{r} - \mathbf{r}')$ and the solution of the Schrödinger equation $\psi^+(\mathbf{r}; k)$ for the outgoing wave boundary condition at the surface of the cluster [22]

$$\begin{aligned} & \int_{S_o} [G_0^+(\mathbf{r}'_o - \mathbf{r}_o; \kappa) \nabla \psi_o^+(\mathbf{r}_o; k) - \psi_o^+(\mathbf{r}_o; k) \\ & \quad \times \nabla G_0^+(\mathbf{r}'_o - \mathbf{r}_o; \kappa)] \cdot \hat{\mathbf{n}}_o d\sigma_o \\ & = \sum_j \int_{S_j} [G_0^+(\mathbf{r}'_j - \mathbf{r}_j; \kappa) \nabla \psi_j^+(\mathbf{r}_j; k) - \psi_j^+(\mathbf{r}_j; k) \\ & \quad \times \nabla G_0^+(\mathbf{r}'_j - \mathbf{r}_j; \kappa)] \cdot \hat{\mathbf{n}}_j d\sigma_j \end{aligned} \quad (1)$$

where $\int \hat{\mathbf{n}}_o d\sigma_o$ and $\int \hat{\mathbf{n}}_j d\sigma_j$ are integrals over the surface of the cluster and of cell j , respectively. A schematic view of the space partitioning of the solution $\psi^+(\mathbf{r}; k)$ is shown in figure 1. In our notations, $k = \sqrt{E}$ is the momentum of the photoelectron, E is the energy of the photoelectron measured from the vacuum level, and $\kappa \equiv \sqrt{k^2 - V_0}$, where V_0 is a constant potential in the interstitial region. In this paper, we adopt the condition $\kappa = k = \sqrt{E - V_0}$ without the outer sphere. For $\kappa = k$, the wavefunction $\psi^+(\mathbf{r}; k)$ satisfies the Lippmann–Schwinger equation

$$\psi^+(\mathbf{r}; k) = \phi_0(\mathbf{r}; k) + \int d\mathbf{r}' G_0^+(\mathbf{r} - \mathbf{r}'; k) V(\mathbf{r}') \psi^+(\mathbf{r}'; k) \quad (2)$$

with $\phi_0(\mathbf{r}; k)$ the free solution and $V(\mathbf{r})$ the potential. We assume that the wavefunctions $\psi_o^+(\mathbf{r}_o; k)$ and $\psi_j^+(\mathbf{r}_j; k)$, respectively, outside and inside of the cluster, can be expanded into local solutions $\Phi_L(\mathbf{r}_o; k)$ and $\Phi_L(\mathbf{r}_j; k)$ as

$$\psi_j^+(\mathbf{r}_j; k) = \sum_L A_L^j(\mathbf{k}) \Phi_L(\mathbf{r}_j; k) \quad (3)$$

$$\psi_o^+(\mathbf{r}_o; k) = \sum_L \tilde{A}_L^o(\mathbf{k}) J_L(\mathbf{r}_o; k) + \sum_L A_L^o(\mathbf{k}) \Phi_L(\mathbf{r}_o; k) \quad (4)$$

where $\tilde{A}_L^o(\mathbf{k}) \equiv i^l \mathcal{Y}_L(\hat{\mathbf{k}}) \sqrt{k/\pi}$. We employ the notation $L = (l, m)$ for the set of the angular azimuthal quantum number l and magnetic quantum number m . The $\mathcal{Y}_L$ are the real spherical harmonics defined as a linear combination of the standard complex spherical harmonics Y_L [30, 31] as

$$\mathcal{Y}_L(\hat{\mathbf{k}}) = \sum_{m'} C_{mm'}^L Y_{lm'}(\hat{\mathbf{k}}) \quad (5)$$

where $C_{mm'}$ for $m > 0$ are

$$\begin{aligned} C_{mm}^l &= (-1)^m / \sqrt{2}, & C_{m-m}^l &= 1 / \sqrt{2} \\ C_{-mm}^l &= -i(-1)^m / \sqrt{2}, & C_{-m-m}^l &= i / \sqrt{2}, \end{aligned}$$

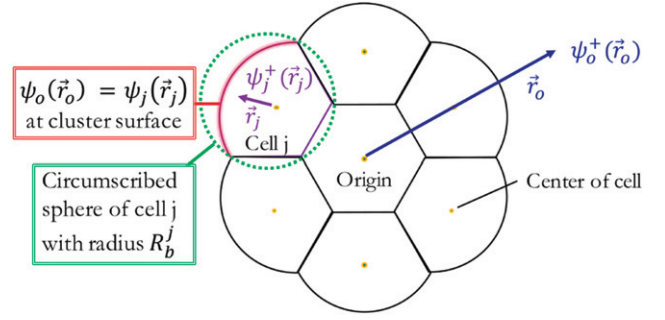


Figure 1. The wavefunction $\psi_j^+(\mathbf{r}_j)$ defined in cell j inside the cluster, and the wavefunction $\psi_o^+(\mathbf{r}_o)$ for the outer region of the cluster, where we omit k .

and $C_{00}^l = 1$, otherwise $C_{mm'}^l = 0$. The functions $J_L(\mathbf{r}; k)$ and $\tilde{H}_L^+(\mathbf{r}; k)$ are defined as $J_L(\mathbf{r}; k) \equiv j_l(kr) \mathcal{Y}_L(\hat{\mathbf{r}})$ and $\tilde{H}_L^+(\mathbf{r}; k) \equiv -ik h_l(kr) \mathcal{Y}_L(\hat{\mathbf{r}})$, where j_l and h_l are, respectively, the spherical Bessel function and the spherical Hankel function of the first kind of order l . The expansion of the local solution into real spherical harmonics is

$$\Phi_L(\mathbf{r}_j; k) = \sum_{L'} R_{L'L}^j(r_j; k) \mathcal{Y}_{L'}(\hat{\mathbf{r}}_j). \quad (6)$$

To obtain the radial solution $R_{L'L}^j$, we employ the 3D Numerov method [22] to solve the three dimensional local Schrödinger equation. This avoids expanding the truncated potential into spherical harmonics, which is known to lead to Gibbs-type [32] convergence problems.

The matrix elements of the free electron Green's function in equation (1) expanded into real spherical harmonics on each site are called the real-space KKR structure factors and are given by [33]

$$G_{LL'}^{ij} \equiv 4\pi \sum_{L''} i^{l-l'+l''} C(L, L', L'') \tilde{H}_{L''}^+(\mathbf{R}_{ij}; k), \quad (7)$$

where $C(L, L', L'') \equiv \int d\hat{\mathbf{r}} \mathcal{Y}_L(\hat{\mathbf{r}}) \mathcal{Y}_{L'}(\hat{\mathbf{r}}) \mathcal{Y}_{L''}(\hat{\mathbf{r}})$ is a Gaunt coefficient of the real spherical harmonics.

On the basis of Green's theorem, the sum of the surface integrals on the truncated cells of the cluster in equation (1) is equal to the surface integral on its bounding sphere (the equality is also valid if $V_0 \neq 0$, as shown in [22]).

Equation (1) can then be reformulated in terms of the matrices S and E (which asymptotically behave as the sine and exponential function, respectively) [34]

$$S_{L'L}^i \equiv (R_b^i)^2 \mathcal{W}[j_L(kr_i), R_{L'L}^i(r_i; k)]_{r_i=R_b^i}, \quad (8)$$

$$E_{L'L}^i \equiv (R_b^i)^2 \mathcal{W}[-ik h_L(kr_i), R_{L'L}^i(r_i; k)]_{r_i=R_b^i}. \quad (9)$$

Here, $\mathcal{W}$ is the Wronskian at the surface of the bounding sphere of scattering site i , and R_b^i is the corresponding radius (see figure 1).

At this stage, we introduce the transition operator (or scattering T matrix) defined by

$$T^i \equiv -S^i (E^i)^{-1} \quad (10)$$

and the coefficients B_L^i related to the coefficients A_L^i by

$$B_L^i(\mathbf{k}) \equiv \sum_{L'} S_{LL'}^i A_{L'}^i(\mathbf{k}). \quad (11)$$

Inserting equations (3) and (7) in the equality equation (1) and using equations (10) and (11), we obtain the following expression for the wavefunction,

$$\psi_i^+(\mathbf{r}_i; k) = \sum_L B_L^i(\mathbf{k}) \bar{\Phi}_L(\mathbf{r}_i) \quad (12)$$

where

$$B_L^i(\mathbf{k}) = \sum_{jL'} \tau_{LL'}^{ij} I_{L'}^j(\mathbf{k}), \quad (13)$$

$$\bar{\Phi}_L(\mathbf{r}_i) = \sum_{L'} ({}^tS)_{LL'}^{-1} \Phi_{L'}(\mathbf{r}_i), \quad (14)$$

$$\tau \equiv (T^{-1} - G)^{-1}, \quad (15)$$

$$I_L^i(\mathbf{k}) \equiv i^l \sqrt{\frac{k}{\pi}} e^{i\mathbf{k} \cdot \mathbf{R}_{io}} \mathcal{Y}_L(\hat{\mathbf{k}}). \quad (16)$$

Here, tS denotes the transpose of the matrix S . The amplitudes $B_L^i(\mathbf{k})$ are determined by the constraint that the wavefunction and its first derivative match smoothly to the outer multiple scattering scheme at the surface at each cell. The multiple scattering matrix τ (or scattering path operator) is constructed from the transition operator and free electron Green's function defined in equation (7).

2.2. Expression of photoelectron diffraction cross-section

In this subsection, we study the PED cross-section based on Fermi's golden rule within the single channel approximation. We assume that the wavefunctions of the core electron, photoelectron and valence electrons do not mix-up and can be treated independently. Under the conditions mentioned above and within the electric dipole approximation, the many-body electric dipole matrix element reduces to a single particle one. Then, the general expression of the PED cross-section along direction $\mathbf{k}$ from an emitter atom on site i is given by

$$I(\mathbf{k}, \hat{\varepsilon}) = 8\pi^2 \alpha \hbar \omega \sum_{m_c} \left| \langle \psi_i^-(\mathbf{r}_i; \mathbf{k}) | \hat{\varepsilon} \cdot \mathbf{r}_i | \phi_{L_c}^c(\mathbf{r}_i) \rangle \right|^2, \quad (17)$$

where $\phi_{L_c}^c(\mathbf{r})$ with the short-hand notation $L_c = (l_c, m_c)$ is the initial core state, α is the fine structure constant, $\hbar \omega$ is the photon energy and $\hat{\varepsilon}$ is the polarization vector of the photon. We neglect the spin polarization in the photoemission process, therefore introducing a factor of two into the formula above.

Using the multiple scattering expression for the photoelectron wavefunction $\psi_i^+(\mathbf{r}_i; \mathbf{k})$ in equation (12), we obtain the following formula for the MFPADs

$$I(\mathbf{k}, \hat{\varepsilon}) = 8\pi^2 \alpha \hbar \omega \sum_{m_c} \left| \sum_L B_L^{i*}(\mathbf{k}) M_{L_c L}(\hat{\varepsilon}) \right|^2, \quad (18)$$

where

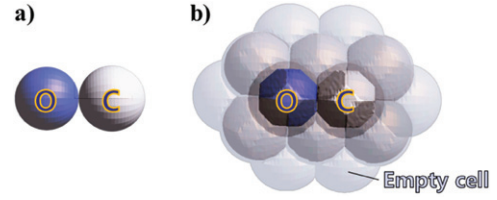


Figure 2. Cell images of CO used for our multiple scattering calculations, (a) for the muffin-tin approximation with two spherical non-overlapping atomic spheres and (b) for the full-potential method with two Voronoi atomic cells, and 22 truncated spherical cells (the so-called empty cells), which contain only electron charge density (no nucleus inside). The positions of cells are arranged in a bcc structure.

$$M_{L_c L}(\hat{\varepsilon}) \equiv \int d\mathbf{r} \bar{\Phi}_L(\mathbf{r}) \hat{\varepsilon} \cdot \mathbf{r} \phi_{L_c}^c(\mathbf{r}),$$

is the transition matrix which describes the excitation by x-rays of polarization $\hat{\varepsilon}$ of an electron from a core orbital $\phi_{L_c}^c$.

2.3. Polarization average of photoelectron cross-section

The polarization average of the MFPADs is obtained by angular integration of the polarization

$$\begin{aligned} \langle I(\mathbf{k}) \rangle_{\varepsilon} &\equiv \frac{1}{4\pi} \int d\hat{\varepsilon} I(\mathbf{k}, \hat{\varepsilon}) \\ &= 2\pi \alpha \hbar \omega \sum_{m_c L L'} B_L^{i*}(\mathbf{k}) B_{L'}^i(\mathbf{k}) \int d\hat{\varepsilon} M_{L_c L}(\hat{\varepsilon}) M_{L_c L'}^*(\hat{\varepsilon}). \end{aligned}$$

Since the electric dipole operator can be expanded as $\hat{\varepsilon} \cdot \mathbf{r} = (4\pi/3) r \sum_m \mathcal{Y}_{1m}(\hat{\varepsilon}) \mathcal{Y}_{1m}(\hat{\mathbf{r}})$ [35], the integration reduces to a sum of Cartesian components,

$$\begin{aligned} &\int d\hat{\varepsilon} M_{L_c L}(\hat{\varepsilon}) M_{L_c L'}^*(\hat{\varepsilon}) \\ &= \frac{4\pi}{3} \left[M_{L_c L}(\hat{\mathbf{x}}) M_{L_c L'}^*(\hat{\mathbf{x}}) + M_{L_c L}(\hat{\mathbf{y}}) M_{L_c L'}^*(\hat{\mathbf{y}}) + M_{L_c L}(\hat{\mathbf{z}}) M_{L_c L'}^*(\hat{\mathbf{z}}) \right]. \end{aligned}$$

Thus, we obtain the following expression for the PA-MFPADs

$$\langle I(\mathbf{k}) \rangle_{\varepsilon} = \frac{1}{3} \left[I(\mathbf{k}, \hat{\mathbf{x}}) + I(\mathbf{k}, \hat{\mathbf{y}}) + I(\mathbf{k}, \hat{\mathbf{z}}) \right].$$

We see that averaged intensity of the MFPADs over the polarization vector along the x , y and z -axes reduces to the intensity of the PA-MFPADs.

The intensity of the PA-MFPADs denoted as $\langle I(\mathbf{k}) \rangle_{\varepsilon}$ is given as

$$\begin{aligned} \langle I(\mathbf{k}) \rangle_{\varepsilon} &= \frac{8\pi^2 \alpha \hbar \omega}{3} \sum_{n, m_c} \left| \sqrt{\frac{4\pi}{3}} \sum_{L L'} B_L^{i*}(\mathbf{k}) \right. \\ &\quad \times C(L_c, 1n, L') \int dr r^3 R_{L' L}(r; k) R_{L_c}^c(r) \left. \right|^2. \end{aligned} \quad (19)$$

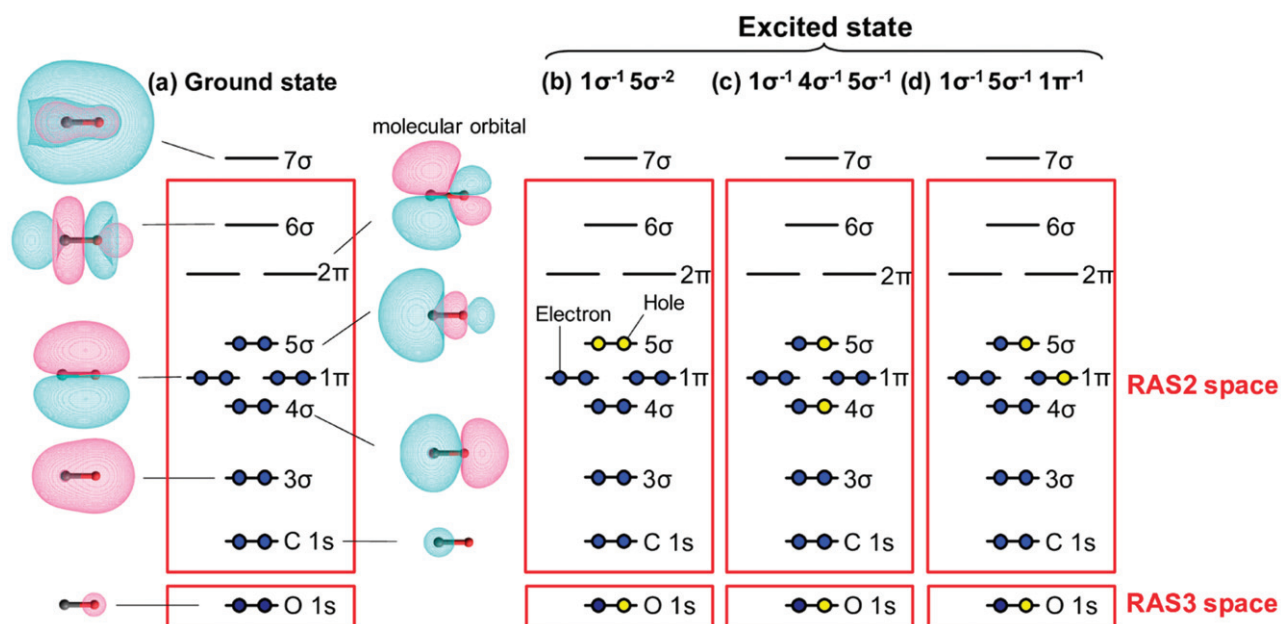


Figure 3. Major electronic configurations of (a) the ground state and the most probable excited states after oxygen $1s$ excitation, (b) $1\sigma^{-1}5\sigma^{-2}$, (c) $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and (d) $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$, respectively. The excited states (b)–(d) are the dicationic states populated via the $C(1s)$ KVV Auger decay with a core hole created by the second pulse in the oxygen $1s$ orbital, that is to say CO^{3+} . Blue and yellow dots in the energy levels represent, respectively, electrons and holes. The shapes of the molecular orbitals are illustrated. Active spaces (RAS2 and RAS3 space) for RASSCF calculations are assigned in the red boxes.

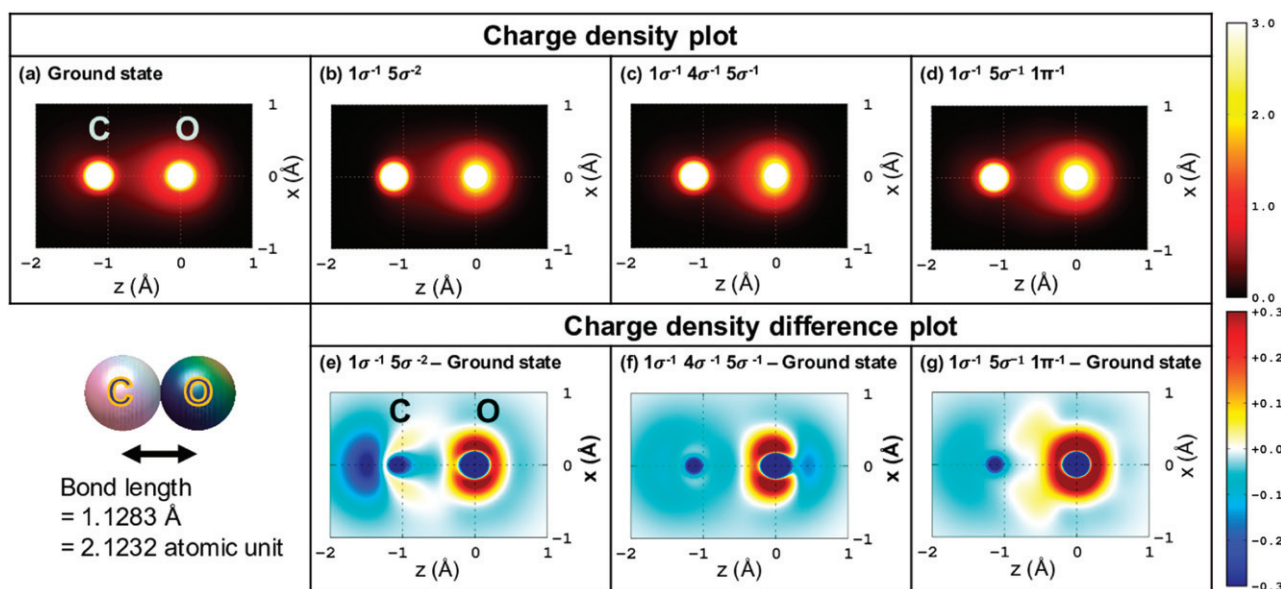


Figure 4. Electron charge density plots of CO in the molecular axis plane for (a) the ground state and the excited states, (b) $1\sigma^{-1}5\sigma^{-2}$, (c) $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and (d) $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$ defined in figure 3. These are the dicationic states populated via the $C(1s)$ KVV Auger decay with a core hole created by the second pulse in the oxygen $1s$ orbital, that is to say CO^{3+} . The electronic structures are calculated with the ANO-RCC-VQZP basis set, where the oxygen atom is set at the origin and the C–O bond length is fixed to 1.1283 Å for all the calculations. The oxygen $1s$ hole in the RAS3 space is kept during the calculations for the excited states. The plots (e)–(g) are electron charge density difference plots between (a) the ground state and the corresponding excited state among (b)–(d), respectively.

3. Computational details

3.1. Quantum chemistry calculation of the electronic structure

In the following, we specialize in a two-color XFEL pump–probe experiment. The pump pulse removes an electron

from the 2σ ($C(1s)$) orbital of CO which results in the subsequent Auger ionization to, e.g. the singlet $5\sigma^{-2}$ states [36–38]. The probe pulse further eliminates an electron from the $1\sigma^{-1}$ ($O(1s)$) orbital. This pump–probe process finally produces a doublet $1\sigma^{-1}5\sigma^{-2}$ state. All electronic structure calculations in CO were carried out with the MOLCAS 8.2 quantum

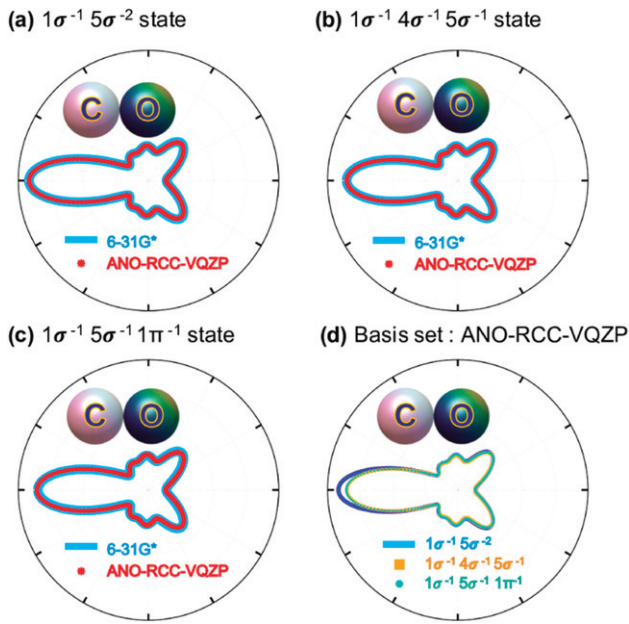


Figure 5. The influence of the basis functions on the PA-MFPADs at 100 eV kinetic energy of the photoelectron with 1.1283 Å of C–O bond length (equilibrium bond length in the ground state). The calculations are performed within the full-potential method using 6-31G* and the ANO-RCC-VQZP basis sets. (a) The $1\sigma^{-1}5\sigma^{-2}$, (b) $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and (c) $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$ states, respectively. These are the dicationic states populated via the C(1s) KVV Auger decay with a core hole created by the second pulse in the oxygen 1s orbital, that is to say CO^{3+} . (d) The influence of the electronic structure type on the PA-MFPADs. We compare the PA-MFPADs for the following electronic structures: $1\sigma^{-1}5\sigma^{-2}$, $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$ states, where the basis set is ANO-RCC-VQZP.

chemical package [39]. The electrostatic potential of the doublet $1\sigma^{-1}5\sigma^{-2}$ state of CO^{3+} as a function of C–O bond length for the multiple scattering calculation were constructed at the multireference second-order perturbation theory with a restricted active space (RASPT2) level of theory [40, 41]. An ionization potential electron affinity shift of 0.25 a.u. was introduced to reduce the intruder problem [42]. The reference wavefunctions for the RASPT2 calculations were computed at the state-averaged restricted active space self-consistent field (SA-RASSCF) level of theory [43]. The ten low-lying doublet states were considered without symmetry, and their average energy was minimized with the equal weight. For all RASPT2 and SA-RASSCF calculations, we included ten spatial orbitals constructed mainly from $1s$, $2s$, $2p$ orbitals of carbon and oxygen atoms, that is 1σ , 2σ , 3σ , 4σ , $2 \times 1\pi$, 5σ (HOMO), $2 \times 2\pi$ (LUMO) and 6σ as active space, where the maximum occupation of the 1σ orbital is restricted to one to describe the core hole in this orbital. The 11 electrons in the active orbitals were treated as active electrons. The ANO-RCC-VQZP basis set [44] was consistently used. Relativistic effects were taken into account through the second order Douglas–Kroll–Hess [45] and mean-field spin–orbit [46] Hamiltonians.

This RASPT2 treatment is essential for an accurate evaluation of the chemical shift in photoelectron energy and photoelectron momentum. The second-order perturbation and relativistic corrections affect only the energetics in this

system. The core orbital is identical to the O(1s) HF orbital of CO^{2+} . We used the natural orbitals obtained from RASPT2 calculations for the input of the electron density calculations; however, the RASPT2 natural orbitals are almost identical to those from RASSCF. Thus, the RASPT2 density is almost identical to the RASSCF one, and the electron density triple cation is close to that of an HF calculation on the ion, especially for the $1\sigma^{-1}5\sigma^{-2}$ state in the Franck–Condon region. We also calculated the $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$ states for comparison. However, they have multiconfiguration characters and RASSCF (or RASPT2) density is essentially different from HF ones. In addition, the contribution of the $1\sigma^{-1}5\sigma^{-2}$ configuration on the RASSCF wavefunction varies when we elongate the C–O bond length. This is another reason that the RASSCF (or RASPT2) treatment is essential.

3.2. Multiple scattering calculation for full-potential and muffin-tin cases

The scattering matrix τ and the transition matrices T were calculated with the FPMS code [21, 22] within the framework of the multiple scattering approach. The PED calculations were performed with the MsSpec code [47], using these matrices as an input. For full-potential calculations, we need to add empty cells (pseudo atomic cells with electron charge density but no nucleus) to account for the charge density outside the (Voronoi) atomic cells. We have used 22 empty cells to embed the carbon and oxygen cells within a bcc structure, as shown in figure 2. The multiple scattering calculation was found to converge at $l_{\max} = 6$ and 8 for the muffin-tin approximation and the full-potential method, respectively. The same $l_{\max}$ was used for the expansion of the wavefunction $\psi_j^+(\mathbf{r}; k)$ into local solutions and the expansion of each local solution into spherical harmonics. Following Kazama *et al* [48], the kinetic energy of the photoelectron was estimated as $k = \sqrt{E - V_0}$, where V_0 is the surface average of the potential of the cluster. This V_0 is the threshold for the continuum and bound state. The wavefunction outside the cells is a plane wave of momentum k .

To estimate the T -matrices for each cell, we solved the local Schrödinger equations with an optical potential, which is the sum of the static Coulomb potential, namely the Hartree potential, and the exchange–correlation potential. To take into account the electron correlation between the ejected core electron and the $N - 1$ remaining electrons, we used a Hedin–Lundqvist potential [49, 50], which is a self-energy used in the field of solid state physics. This potential is based on the GW approximation with a non-SCF Green’s function G , a screened potential W and a single plasmon pole approximation, and allows one to satisfactorily model the scattering optical potential in the multiple scattering calculations. It is known, however, that its imaginary part produces a rather strong damping effect [21, 22], especially for light elements such as carbon and oxygen; hence, we used only the real part in our calculations. Since CO molecules do not have any particular direction perpendicular to the molecular axis, the intensity of the photoelectron was averaged over the azimuth angle around that axis.

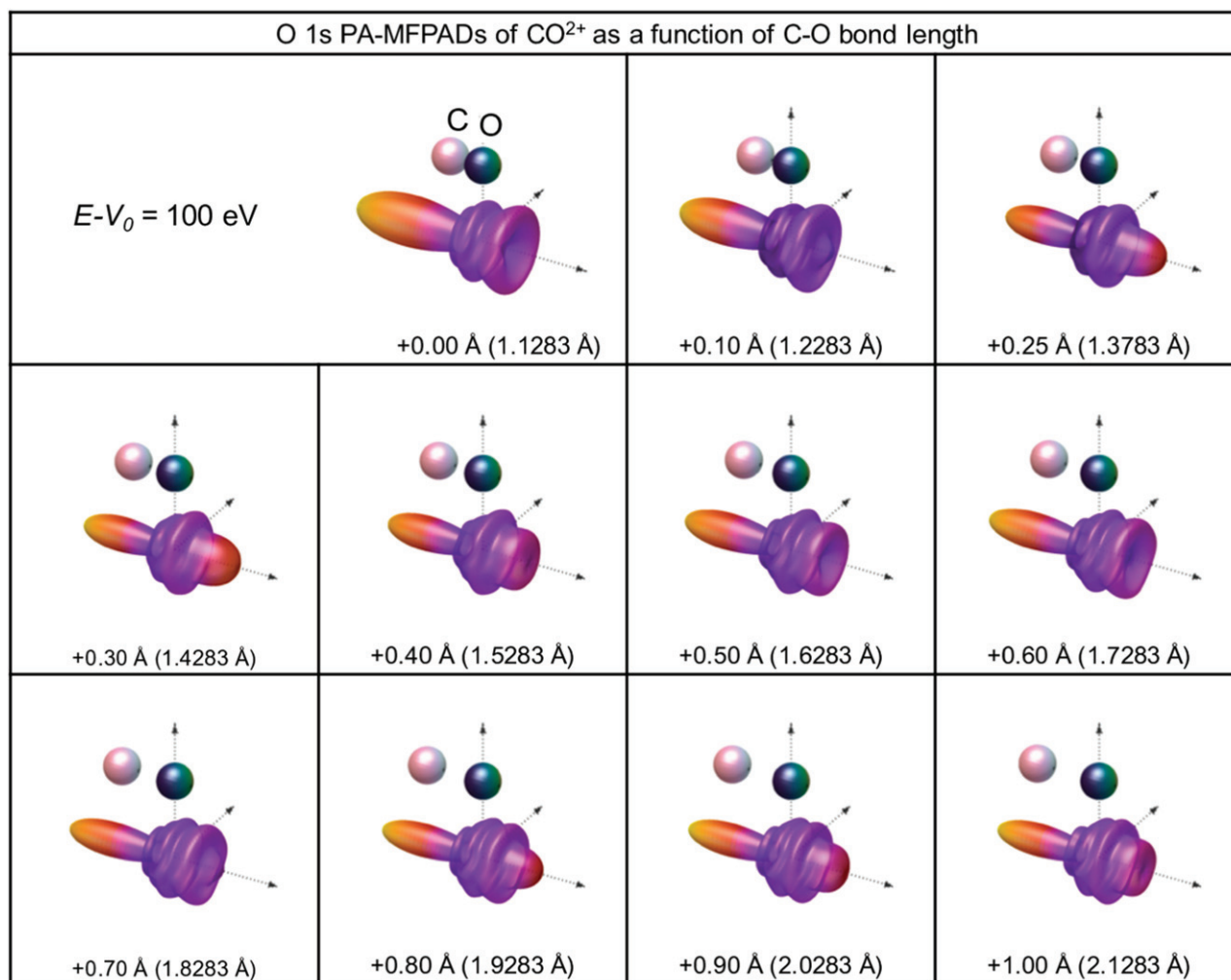


Figure 6. The oxygen 1s PA-MFPADs of CO²⁺ as a function of the C–O bond length R from 1.1283 Å (equilibrium bond length in ground state) to 2.1283 Å. All the calculations were performed within the full-potential method for the excited state $1\sigma^{-1}5\sigma^{-2}$ (i.e. O $1s^{-1}$ HOMO⁻², which is triply charged CO, namely CO³⁺) with a photoelectron kinetic energy of 100 eV.

The electron charge density for the multiple scattering calculations has been prepared following two different prescriptions. In the first, which we refer to as ‘non-SCF’ (non-self-consistent), the molecular potential and the electron charge density were constructed by a simple superposition of the free atomic densities. The free-atom calculations were performed using the Desclaux code [51], which employs a multi-configuration relativistic Dirac–Fock theory, and the excited states were calculated using a fully relaxed core hole. The second one, which we call ‘SCF’, employs the electron charge density calculated from MOLCAS 8.2 with the help of a Hermite Gaussian basis function set [52] and the (Coulomb) static part of the scattering potential calculated under the McMurchie–Davidson scheme [52] with a Boys function [53]. The details can be found in [54].

In the muffin-tin calculations, for both prescriptions, the electron charge density and the static part of the potential were spherically averaged. For the muffin-tin spheres, we used touching but non-overlapping C and O atomic spheres. The sphere radii were estimated for each bond length by the Norman criterion [55] and are roughly proportional to the

ionic radius of the atom in the periodic table. An empirical prescription to improve the results uses overlapping spheres, 10% ~ 15%, to account, approximately, for the presence of covalent electrons (see [55, 56]). This approximation is called the atomic-sphere approximation in LMTO and KKR [57]. However, as it is not our scope in this work to empirically optimise the results through tentative overlapping, we have kept to the touching spheres. For V_0 , we have used the value obtained from our full-potential calculations.

4. Results

4.1. Dependence on electronic structures

We consider three states, $1\sigma^{-1}5\sigma^{-2}$, $1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ and $1\sigma^{-1}5\sigma^{-1}1\pi^{-1}$, as shown in figure 3. These states are the dicationic states populated via the C(1s) KVV Auger decay [38] with a core hole created by the second pulse in the oxygen 1s orbital.

In figure 4 the ground state and excited state electron charge densities are displayed in the upper row, and the electron charge density difference plots, between each excited state and the ground state, are shown in the lower row. As

$1\sigma^{-1}4\sigma^{-1}5\sigma^{-1}$ in (c) has a C_{2v} symmetry around the molecular axis, we chose the xz -plane to lay the orbital in order to see its anisotropy. All the other electronic configurations have a $C_{\infty v}$ symmetry around the axis. These comparisons show the change in charge distributions varying with the electronic configuration.

Next, we studied the influence of the electronic structure on the PA-MFPADs. Figure 5 shows the results of PA-MFPADs simulations performed for the three electron configurations using the 6-31G* (small, double-zeta + valence polarization) and the ANO-RCC-VQZP (large, quadratic-zeta + valence polarization) basis sets. The results clearly indicate that the PA-MFPADs are not very sensitive to the electronic structure in this high-energy regime. Therefore, in the following subsections, we will use the $1\sigma^{-1}5\sigma^{-2}$ configuration as a representative electronic configuration using the ANO-RCC-VQZP.

4.2. Dependence on bond length

The PA-MFPADs calculated as a function of the C–O bond length are shown in figure 6. The forward-intensity $\langle I(k, \theta = 0^\circ) \rangle_\epsilon$ corresponds to the largest lobe, which is directed along the bond toward the carbon atom. This lobe is due to the so-called focusing effect of photoelectrons by the carbon atom. The backward-intensity $\langle I(k, \theta = 180^\circ) \rangle_\epsilon$ is in the opposite direction, and it oscillates between the peak top and the valley of the lobe as a function of R . Within the framework of the plane wave single-scattering approximation of the electron Green's function in equation (7) (see for instance the Extended X-Ray Absorption Fine Structure (EXAFS) formula [58]), one period of the oscillation corresponds to an increase by $\Delta R = \pi/k$ in R . There are several small lobes between the forward and backward directions, which move from backward to forward as R increases. The number of lobes is incremented every one period of oscillation of the backward-intensity peak. Labeling angles θ_1 and θ_2 for two neighboring lobes ($\theta_1 < \theta_2$), we have approximately $R = 2\pi/k(\cos \theta_1 - \cos \theta_2)$. Thus, the PA-MFPADs do contain the information about the bond length. We will discuss this relation in more detail in the companion paper [59].

4.3. Dependence on shape of potential

To evaluate the effect of the muffin-tin approximation on the PA-MFPADs, we compare in figure 7 the oxygen $1s$ PA-MFPADs of CO^{2+} calculated using the muffin-tin approximation to the corresponding one obtained from the full-potential method. The major C($1s$) KVV Auger decay channels in the low-energy region were estimated according to the ADC(2) calculations by Cederbaum *et al* [38]. The kinetic energy of the photoelectron is chosen to be 100 eV on the basis of the widely supported idea that photoelectron spectroscopy is less sensitive to the details of the electronic structure and the electron charge density in a sufficiently high-energy regime ($\gtrsim 100$ eV) [22].⁷

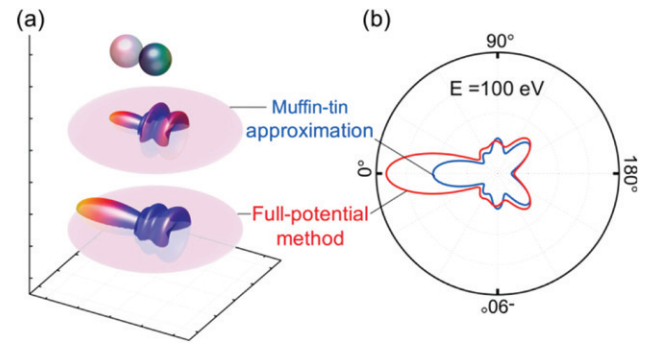


Figure 7. The oxygen $1s$ PA-MFPADs of CO^{2+} in the excited state $1\sigma^{-1}5\sigma^{-2}$ (i.e. $\text{O } 1s^{-1} \text{ HOMO}^{-2}$, which is triply charged CO, namely CO^{3+}) with 100 eV of photoelectron kinetic energy. The oxygen atom is located at the origin, and the azimuthal angle is defined from the direction of the carbon atom. (a) The oxygen $1s$ PA-MFPADs within the muffin-tin approximation is shown in the upper plane and that within the full-potential method is displayed in the lower plane. (b) Comparison of the PA-MFPADs between the muffin-tin approximation (blue line) and the full-potential method (red line).

We see that the PA-MFPAD patterns are clearly distinct from each other. The largest contribution to the forward intensity comes from the forward scattering by the carbon atom, which induces a focusing effect. This forward scattering is due here to a weak potential. Since such a weak potential is essentially left outside of the muffin-tin spheres in the case of short distances, the forward intensity is certainly underestimated in the muffin-tin approximation. In figure 8, we show a series of comparisons between the muffin-tin approximation and the full-potential method for multiple scattering calculations of the oxygen $1s$ PA-MFPADs of CO^{2+} as a function of the C–O bond length R from 1.1283 Å (equilibrium bond length in the ground state) to 2.1283 Å. All the calculations were performed for the excited state $1\sigma^{-1}5\sigma^{-2}$ (i.e. $\text{O } 1s^{-1} \text{ HOMO}^{-2}$) with a photoelectron kinetic energy of 100 eV. As noted in figure 7, the forward-intensity peak is significantly underestimated by the muffin-tin approximation at near equilibrium distance. Moreover, the peak positions between the forward- and backward-intensity peaks start to differ with the increase in R between the results in the muffin-tin approximation and the full-potential method.

As we observe in figure 9, the frequency of the oscillations of the backward intensity is almost $2kR$ for both cases. Indeed, this corresponds exactly to the EXAFS oscillation frequency. In this high-energy regime, i.e. in the EXAFS regime, single scattering toward the backward direction is the main contribution of the backward intensity, so that we expect that we may observe this EXAFS-like oscillation in the backward intensity as in EXAFS. In the oxygen $1s$ PA-MFPADs of CO^{2+} , the back-scattering is caused mainly by the strong (core) potential very near the center of the carbon atom. At first sight, it may seem strange to have so much difference between the two calculations, since the electron charge densities are almost spherically symmetric in the full-potential case. In fact, it is the contribution of the forward scattering by the oxygen atom which accounts for the difference. This forward scattering occurs at the second scattering order, along the pathway going from the

⁷ The calculated vertical $\text{O}(1s^{-1})$ ionization energy from the lowest singlet state of CO^{2+} was found to be 564–569 eV at the RASPT2/ANO-RCC-VQZP level of theory depending on the C–O bond length R (chemical shifts). This means that we have only 5% and 2%–3% errors in the photoelectron energy and momentum, respectively.

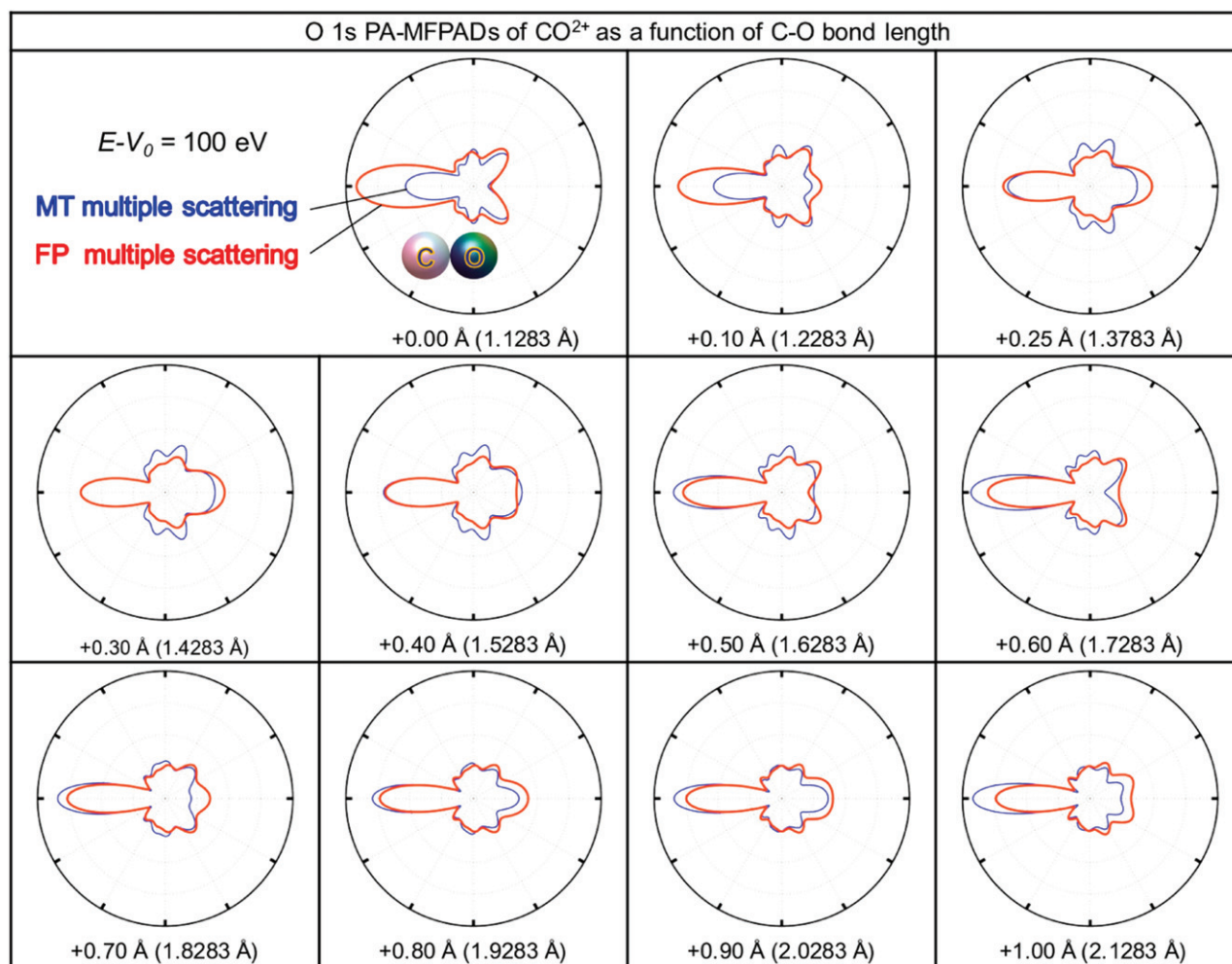


Figure 8. The oxygen 1s PA-MFPADs of CO^{2+} as a function of the C–O bond length R from 1.1283 Å (equilibrium bond length in the ground state) to 2.1283 Å for a photoelectron kinetic energy of 100 eV. All the calculations were performed with the excited state $1\sigma^{-1}5\sigma^{-2}$ (i.e. $\text{O } 1s^{-1}$ HOMO $^{-2}$, which is triply charged CO, namely CO^{3+}). Blue and red lines indicate the results of multiple scattering calculations using the muffin-tin approximation and the full-potential method, respectively.

carbon atom to the detector. Its influence, especially through the phase of the forward scattering amplitude, $\psi(k, \theta = 0^\circ)$, appears in the EXAFS oscillation as $\cos(2kR + \psi(k, \theta = 0^\circ) + \psi(k, \theta = 180^\circ))$. Since forward scattering is important even for a weak potential, it is more sensitive to fine details of the electron charge density and, therefore, it depends strongly on the modeling of the scattering potential. Thus the error in the muffin-tin approximation appears to make the difference in the peak/valley positions for the two calculations in figure 9.

4.4. Toward reaction dynamics imaging using PA-MFPADs based on *ab-initio* full-potential multiple scattering theory

Finally, we would like to consider the potential targets of the reaction dynamics imaging simulations using PA-MFPADs based on the full-potential multiple scattering theory, just beyond the dissociation of the dication of diatomic molecules.

A possible extension that may be monitored by the present technique is the fragmentation of the methanol dication. There are at least two major fragmentation channels. One is the direct C–O bond breaking [60]. This process is very similar to the

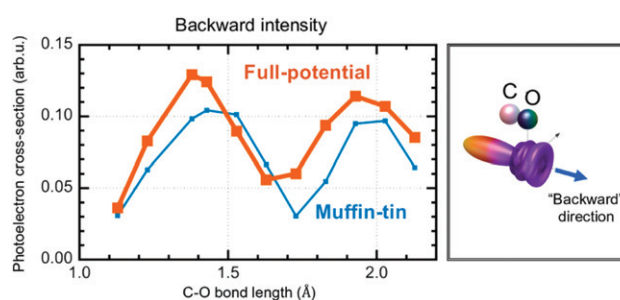


Figure 9. Backward intensities of the oxygen 1s PA-MFPADs of CO^{2+} as a function of the C–O bond length R from 1.1283 Å (equilibrium bond length in the ground state) to 2.1283 Å for the $1\sigma^{-1}5\sigma^{-2}$ state (i.e. $\text{O } 1s^{-1}$ HOMO $^{-2}$, which is triply charged CO, namely CO^{3+}). Dark blue and orange lines indicate multiple scattering calculation results within the muffin-tin approximation and the full-potential method, respectively.

dissociation of the diatomic molecule, which we have considered in the present work. The other is the molecular dissociation into CO^+ and H_3^+ . In this process, roaming of H_2 takes place and extraction of H^+ from CH or OH sites follows [61].

The timescale of this process is ~ 100 fs. We should be able to distinguish between these two pathways, both experimentally and theoretically, by the momenta of fragment ions that define the molecular axis or reaction plane to be observed for O 1s PA-MFPADs via the C(1s) pump O(1s) probe scheme.

A case a little more complex but still feasible to the present method could be the x-ray induced fragmentation of doubly charged amino acids such as glycine [62] and serine [63]. A typical first step of the decomposition is the COOH^+ ejection [63–65] with a time scale of 10–100 fs [62–65]. Capturing the COOH^+ loss dynamics could be achieved by the C(1s) pump N(1s) probe scheme.

Furthermore, many photochemical reactions of polyatomic molecules with hetero-atoms for the photoelectron source could also become attractive targets. For example, ultraviolet-induced femtosecond H loss of pyrrole via $^1\pi\pi^*$ and $^1\pi\sigma^*$ states [66] will be a suitable target since its N atom is applicable to the photoelectron source and the dissociating H atom locates just next to the N atom. These molecules will also become good potential targets of the imaging experiments with the PA-MFPADs with the two-color XFEL light source.

From the theoretical points of view, the on-the-fly reaction dynamics calculations [60, 67–75] and *ab-initio* core-hole spectroscopy [71, 76–81] for these polyatomic molecules are already feasible. It will also be helpful to improve the efficiency of the electrostatic potential evaluation and the full-potential multiple scattering calculations by further tuning, including massive parallelization, to push its interplay with the experiment and reaction dynamics theory forward. In addition, analytical formulae that extract structural dynamics from the complex scattering results will also be useful [59].

5. Conclusions

We have theoretically studied the PA-MFPADs emitted from the 1s orbital of the oxygen atom of dissociating dicationic carbon monoxide CO^{2+} . Within the framework of the multiple scattering theory, we have investigated the influence of the muffin-tin approximation and the role of the potential in different excited states, with electron charge densities calculated using the RASPT2 method to properly model the influence of the core hole in the final state. The full-potential theory takes into account the electron charge density outside the physical atomic cells by simultaneously using Voronoi polyhedra and truncated spheres.

Despite the fact that the muffin-tin approximation was giving similar results to the full-potential calculation above 100 eV for XANES [22], we observe here significant differences in the PA-MFPADs. This may be explained by the general optical theorem, which states, for *real potentials*, that the integrated photoemission over the whole solid angle equals the x-ray absorption [22, 82]. A consequence of this theorem is that in XANES, even if there is a difference in angular distribution between the muffin-tin approximation and the full-potential method, this difference will be smeared out by the angular integration. Moreover, even in the case of long bond lengths, where the atomic approximation works, we found a

significant difference in the PA-MFPADs between the muffin-tin approximation and the full-potential method. The consequence is that we need to use the full-potential multiple scattering theory to study structure dynamics using FELs and synchrotrons, even in the high-energy region.

We have also studied the PA-MFPADs from different excited states of the Auger final state. We have found that the type of excited states does not significantly affect the PA-MFPADs, though the difference in the electronic structure was not negligible. The relationship between the PA-MFPAD patterns and the C–O bond length will be discussed in detail within the multiple scattering theory in the companion article [59].

Acknowledgment

We would like to acknowledge C R Natoli for fruitful discussions. This work was accomplished under the program of the Dynamic Alliance for Open Innovation Bridging Human, Environment and Materials program and the Cooperative Research Program of the ‘Network Joint Research Center for Materials and Devices’. KH acknowledges funding by JSPS KAKENHI under Grant Nos. 18K05027 and 17K04980. KY is grateful for the financial support from the Building of Consortia for the Development of Human Resources in Science and Technology, MEXT.

ORCID iDs

K Hatada  <https://orcid.org/0000-0002-3745-2014>

References

- [1] Woodruff D P 2008 Photoelectron diffraction: from phenomenological demonstration to practical tool *Appl. Phys. A* **92** 439–45
- [2] Shigemasa E, Adachi J, Oura M and Yagishita A 1995 Angular distributions of 1s photoelectrons from fixed-in-space N_2 molecules *Phys. Rev. Lett.* **74** 359–62
- [3] Jagutzki O *et al* 2002 Multiple hit readout of a microchannel plate detector with a three-layer delay-line anode *IEEE Trans. Nucl. Sci.* **49** 2477–83
- [4] Ullrich J, Moshhammer R, Dorn A, Dörner R, Schmidt L P H and Schmidt-Böcking H 2003 Recoil-ion and electron momentum spectroscopy: reaction-microscopes *Rep. Prog. Phys.* **66** 1463–545
- [5] Landers A *et al* 2001 Photoelectron diffraction mapping: molecules illuminated from within *Phys. Rev. Lett.* **87** 013002
- [6] Poliakoff E D and Lucchese R R 2006 Evolution of photoelectron-vibrational coupling with molecular complexity *Phys. Scr.* **74** C71–9
- [7] Plésiat E, Decleva P and Martín F 2012 Vibrationally resolved photoelectron angular distributions from randomly oriented and fixed-in-space N_2 and CO molecules *J. Phys. B: At. Mol. Opt. Phys.* **45** 194008
- [8] Cherepkov N A, Raseev G, Adachi J, Hikosaka Y, Ito K, Motoki S, Sano M, Soejima K and Yagishita A 2000 K-shell photoionization of CO: II. Determination of dipole matrix elements and phase differences *J. Phys. B: At. Mol. Opt. Phys.* **33** 4213–36

- [9] Kaiser L *et al* 2020 Angular emission distribution of O 1s photoelectrons of uniaxially oriented methanol *J. Phys. B: At. Mol. Opt. Phys.* **53** 194002
- [10] Saito N *et al* 2002 Carbon K-shell photoelectron angular distribution from fixed-in-space CO₂ molecules *J. Phys. B: At. Mol. Opt. Phys.* **36** L25–30
- [11] Rolles D *et al* 2005 Isotope-induced partial localization of core electrons in the homonuclear molecule N₂ *Nature* **437** 711–5
- [12] Semenov S K, Cherepkov N A, Fecher G H and Schönhense G 2000 Generalization of the atomic random-phase-approximation method for diatomic molecules: N₂ photoionization cross-section calculations *Phys. Rev. A* **61** 032704
- [13] Semenov S K and Cherepkov N A 2002 Generalization of atomic random-phase-approximation method for diatomic molecules. II. N₂K-shell photoionization *Phys. Rev. A* **66** 022708
- [14] Stratmann R E and Lucchese R R 1995 A graphical unitary group approach to study multiplet specific multichannel electron correlation effects in the photoionization of O₂ *J. Chem. Phys.* **102** 8493–505
- [15] Stratmann R E, Zuurales R W and Lucchese R R 1996 Multiplet-specific multichannel electron-correlation effects in the photoionization of NO *J. Chem. Phys.* **104** 8989–9000
- [16] Stener M, Fronzoni G and Decleva P 2005 Time-dependent density-functional theory for molecular photoionization with noniterative algorithm and multicenter B-spline basis set: CS₂ and C₆H₆ case studies *J. Chem. Phys.* **122** 234301
- [17] Plésiat E, Decleva P and Martín F 2013 Relationship between polarization-averaged molecular-frame photoelectron angular distributions and geometry *Phys. Rev. A* **88** 063409
- [18] Fukuzawa H *et al* 2019 Probing gaseous molecular structure by molecular-frame photoelectron angular distributions *J. Chem. Phys.* **151** 104302
- [19] Fadley C S *et al* 1994 Photoelectron diffraction and holography: present status and future prospects *J. Electron Spectrosc. Relat. Phenom.* **68** 19–47
- [20] Kazama M, Fujikawa T, Kishimoto N, Mizuno T, Adachi J and Yagishita A 2013 Photoelectron diffraction from single oriented molecules: towards ultrafast structure determination of molecules using x-ray free-electron lasers *Phys. Rev. A* **87** 063417
- [21] Hatada K, Hayakawa K, Benfatto M and Natoli C R 2007 Full-potential multiple scattering for x-ray spectroscopies *Phys. Rev. B* **76** 060102
- [22] Hatada K, Hayakawa K, Benfatto M and Natoli C R 2010 Full-potential multiple scattering theory with space-filling cells for bound and continuum states *J. Phys.: Condens. Matter* **22** 185501
- [23] Xu J, Krüger P, Natoli C R, Hayakawa K, Wu Z and Hatada K 2015 X-ray absorption spectra of graphene and graphene oxide by full-potential multiple scattering calculations with self-consistent charge density *Phys. Rev. B* **92** 125408
- [24] Williams J B *et al* 2012 Imaging polyatomic molecules in three dimensions using molecular frame photoelectron angular distributions *Phys. Rev. Lett.* **108** 233002
- [25] Fukuzawa H *et al* 2019 Probing molecular bond-length using molecular-frame photoelectron angular distributions *J. Chem. Phys.* **150** 174306
- [26] Emma P *et al* 2010 First lasing and operation of an ångström-wavelength free-electron laser *Nat. Photon.* **4** 641–7
- [27] Krasniqi F, Najjari B, Strüder L, Rolles D, Voitkiv A and Ullrich J 2010 Imaging molecules from within: ultrafast angström-scale structure determination of molecules via photoelectron holography using free-electron lasers *Phys. Rev. A* **81** 033411
- [28] Decking W *et al* 2020 A MHz-repetition-rate hard x-ray free-electron laser driven by a superconducting linear accelerator *Nat. Photon.* **14** 391–7
- [29] Kastirke G *et al* 2020 Photoelectron diffraction imaging of a molecular breakup using an x-ray free-electron laser *Phys. Rev. X* **10** 021052
- [30] Gianturco F A and Jain A 1986 The theory of electron scattering from polyatomic molecules *Phys. Rep.* **143** 347–425
- [31] Brouder C, Ruiz López M F, Pettifer R F, Benfatto M and Natoli C R 1989 Systematic approach to the calculation of the polarization-dependent (and polarization-averaged) general term of the curved-wave multiple-scattering series in the x-ray-absorption cross section *Phys. Rev. B* **39** 1488–500
- [32] Williams A R and Morgan J-v-W 1974 Multiple scattering by non-Muffin-tin potentials: general formulation *J. Phys. C: Solid State Phys.* **7** 37–60
- [33] Natoli C R, Benfatto M, Della Longa S and Hatada K 2003 X-ray absorption spectroscopy: state-of-the-art analysis *J. Synchrotron Radiat.* **10** 26–42
- [34] Gonis A and Butler W H 2000 *Multiple Scattering in Solids* (New York: Springer)
- [35] Varshalovich D A, Moskalev A N and Khersonskii V K 1988 *Quantum Theory of Angular Momentum* (Singapore: World Scientific)
- [36] Kelber J A, Jennison D R and Rye R R 1981 Analysis of the Auger spectra of CO and CO₂ *J. Chem. Phys.* **75** 652–62
- [37] Feyer V, Bolognesi P, Coreno M, Prince K C, Avaldi L, Storch L and Tarantelli F 2005 Effects of nuclear dynamics in the low-kinetic-energy Auger spectra of CO and CO₂ *J. Chem. Phys.* **123** 224306
- [38] Cederbaum L S, Campos P, Tarantelli F and Sgamellotti A 1991 Band shape and vibrational structure in Auger spectra: theory and application to carbon monoxide *J. Chem. Phys.* **95** 6634–44
- [39] Aquilante F *et al* 2016 Molcas8: new capabilities for multiconfigurational quantum chemical calculations across the periodic table *J. Comput. Chem.* **37** 506–41
- [40] Malmqvist P Å, Pierloot K, Shahi A R M, Cramer C J and Gagliardi L 2008 The restricted active space followed by second-order perturbation theory method: theory and application to the study of CuO₂ and Cu₂O₂ systems *J. Chem. Phys.* **128** 204109
- [41] Sauri V, Serrano-Andrés L, Shahi A R M, Gagliardi L, Vancoillie S and Pierloot K 2011 Multiconfigurational second-order perturbation theory restricted active space (RASPT2) method for electronic excited states: a benchmark study *J. Chem. Theory Comput.* **7** 153–68
- [42] Ghigo G, Roos B O and Malmqvist P-Å 2004 A modified definition of the zeroth-order Hamiltonian in multiconfigurational perturbation theory (CASPT2) *Chem. Phys. Lett.* **396** 142–9
- [43] Malmqvist P A, Rendell A and Roos B O 1990 The restricted active space self-consistent-field method, implemented with a split graph unitary group approach *J. Phys. Chem.* **94** 5477–82
- [44] Roos B O, Lindh R, Malmqvist P-Å, Veryazov V and Widmark P-O 2004 Main group atoms and dimers studied with a new relativistic ANO basis set *J. Phys. Chem. A* **108** 2851–8
- [45] Peng D and Reiher M 2012 Exact decoupling of the relativistic Fock operator *Theor. Chem. Acc.* **131** 1081
- [46] Heß B A, Marian C M, Wahlgren U and Gropen O 1996 A mean-field spin-orbit method applicable to correlated wavefunctions *Chem. Phys. Lett.* **251** 365–71
- [47] Sébilleau D, Natoli C, Gavaza G M, Zhao H, Da Pieve F and Hatada K 2011 MsSpec-1.0: a multiple scattering package for electron spectroscopies in material science *Comput. Phys. Commun.* **182** 2567–79

- [48] Kazama M, Shinotsuka H, Fujikawa T, Stener M, Decleva P, Adachi J-i, Mizuno T and Yagishita A 2012 Multiple-scattering calculations for 1s photoelectron angular distributions from single oriented molecules in the energy region above 50 eV *J. Electron Spectrosc. Relat. Phenom.* **185** 535–45
- [49] Hedin L and Lundqvist S 1969 Effects of electron–electron and electron–phonon interactions on the one-electron state of solids *Solid State Physics* vol 23 ed F Seitz, D Turnbull and H Ehrenreich (New York: Academic) pp 1–517
- [50] Hedin L and Lundqvist B I 1971 Explicit local exchange-correlation potentials *J. Phys. C: Solid State Phys.* **4** 2064–83
- [51] Desclaux J P 1975 A multiconfiguration relativistic DIRAC–FOCK program *Comput. Phys. Commun.* **9** 31–45
- [52] Helgaker T, Jorgensen P and Olsen J 2000 *Molecular Electronic-Structure Theory* (New York: Wiley)
- [53] Boys S F and Egerton A C 1950 Electronic wave functions: I. A general method of calculation for the stationary states of any molecular system *Proc. R. Soc. A* **200** 542–54
- [54] Komiya N, Ota F, Xu J and Hatada K 2018 ES2MS: interface from electronic structure codes to multiple scattering codes *Multiple Scattering Theory for Spectroscopies* ed D Sébilleau, K Hatada and H Ebert (Berlin: Springer) pp 275–82
- [55] Norman J G Jr 1976 Non-empirical versus empirical choices for overlapping-sphere radii ratios in SCF- X_α -SW calculations on ClO_4^- and SO_2 *Mol. Phys.* **31** 1191–8
- [56] Herman F, Williams A R and Johnson K H 1974 Multiple scattering method based on overlapping atomic spheres with application to the TCNQ molecule *J. Chem. Phys.* **61** 3508–22
- [57] Turek I, Drchal V, Kudrnovsky J, Sob M and Weinberger P 1997 *Electronic Structure of Disordered Alloys, Surfaces and Interfaces* (Berlin: Springer)
- [58] Rehr J J, Albers R C, Natoli C R and Stern E A 1986 New high-energy approximation for x-ray-absorption near-edge structure *Phys. Rev. B* **34** 4350–3
- [59] Ota F, Hatada K, Sébilleau D, Ueda K and Yamazaki K 2020 Theory on polarization-averaged core-level molecular-frame photoelectron angular distributions: II. Extracting the x-ray induced fragmentation dynamics of carbon monoxide dication from forward and backward intensities *J. Phys. B: At. Mol. Opt.* (accepted)
- [60] Zhang Q, Fukahori S, Ando T, Kanya R, Iwasaki A, Rathje T, Paulus G G and Yamanouchi K 2020 Absolute carrier-envelope-phase dependences of single and double ionization of methanol in a near-IR few-cycle laser field *J. Chem. Phys.* **152** 194304
- [61] Ekanayake N *et al* 2017 Mechanisms and time-resolved dynamics for trihydrogen cation (H_3^+) formation from organic molecules in strong laser fields *Sci. Rep.* **7** 4703
- [62] Fang L, Xiong H, Kukuk E, Petrovic V S and Berrah N 2018 X-ray-initiated photodissociation of the glycine molecule *Phys. Rev. A* **98** 053408
- [63] Itälä E, Levola H, Ha D T, Kooser K, Rachlew E and Kukuk E 2016 Photofragmentation of serine following C 1s core ionization-comparison with cysteine *J. Phys. Chem. A* **120** 5419–26
- [64] Ha D T, Wang Y, Alcamí M, Itälä E, Kooser K, Urpelainen S, Huels M A, Kukuk E and Martín F 2014 Fragmentation dynamics of doubly charged methionine in the gas phase *J. Phys. Chem. A* **118** 1374–83
- [65] Ha D T, Yamazaki K, Wang Y, Alcamí M, Maeda S, Kono H, Martín F and Kukuk E 2016 Fragmentation network of doubly charged methionine: interpretation using graph theory *J. Chem. Phys.* **145** 094302
- [66] Montero R, Peralta Conde Á, Ovejas V, Fernández-Fernández M, Castaño F, Vázquez de Aldana J R, Longarte A and Longarte A 2012 Femtosecond evolution of the pyrrole molecule excited in the near part of its UV spectrum *J. Chem. Phys.* **137** 064317
- [67] Yamazaki K, Nakamura T, Niitsu N, Kanno M, Ueda K and Kono H 2014 Communication: two-step explosion processes of highly charged fullerene cations C_{60}^{q+} ($q = 20\text{--}60$) *J. Chem. Phys.* **141** 121105
- [68] Yamazaki K, Niitsu N, Kanno M, Ueda K and Kono H 2019 Capturing the photo-induced dynamics of nano-molecules by x-ray free electron laser induced Coulomb explosion *J. Chem. Phys.* **151** 124305
- [69] Nagaya K *et al* 2016 Femtosecond charge and molecular dynamics of I-containing organic molecules induced by intense x-ray free-electron laser pulses *Faraday Discuss.* **194** 537–62
- [70] Kukuk E *et al* 2015 Internal energy dependence in x-ray-induced molecular fragmentation: an experimental and theoretical study of thiophene *Phys. Rev. A* **91** 043417
- [71] Inhester L *et al* 2018 Chemical understanding of the limited site-specificity in molecular inner-shell photofragmentation *J. Phys. Chem. Lett.* **9** 1156–63
- [72] Rudenko A *et al* 2017 Femtosecond response of polyatomic molecules to ultra-intense hard x-rays *Nature* **546** 129–32
- [73] Richter M, Marquetand P, González-Vázquez J, Sola I and González L 2011 SHARC: *ab initio* molecular dynamics with surface hopping in the adiabatic representation including arbitrary couplings *J. Chem. Theor. Comput.* **7** 1253–8
- [74] Mai S, Marquetand P and González L 2018 Nonadiabatic dynamics: the SHARC approach *Wiley Interdiscip. Rev.-Comput. Mol. Sci.* **8** e1370
- [75] Barbatti M, Ruckebauer M, Plasser F, Pittner J, Granucci G, Persico M and Lischka H 2014 Newton-X: a surface-hopping program for nonadiabatic molecular dynamics *Wiley Interdiscip. Rev.-Comput. Mol. Sci.* **4** 26–33
- [76] Hua W, Mukamel S and Luo Y 2019 Transient x-ray absorption spectral fingerprints of the S_1 dark state in uracil *J. Phys. Chem. Lett.* **10** 7172–8
- [77] Tashiro M, Ueda K and Ehara M 2011 Auger decay of molecular double core-hole state *J. Chem. Phys.* **135** 154307
- [78] Mitani M, Takahashi O, Saito K and Iwata S 2003 Theoretical molecular Auger spectra with electron population analysis *J. Electron Spectrosc. Relat. Phenom.* **128** 103–17
- [79] Khalili K, Inhester L, Arnold C, Gertsen A S, Andreasen J W and Santra R 2020 Simulation of time-resolved x-ray absorption spectroscopy of ultrafast dynamics in particle-hole-excited 4-(2-thienyl)-2, 1, 3-benzothiadiazole *Struct. Dyn.* **7** 044101
- [80] Morzan U N, Videla P E, Soley M B, Nibbering E T J and Batista V S 2020 Vibronic dynamics of photodissociating ICN from simulations of ultrafast x-ray absorption spectroscopy *Angew. Chem., Int. Ed.* **59** 20044–8
- [81] Ehlert C and Klamroth T 2020 PSIXAS: a Psi4 plugin for efficient simulations of x-ray absorption spectra based on the transition-potential and Δ -Kohn–Sham method *J. Comput. Chem.* **41** 1781–9
- [82] Natoli C R, Benfatto M and Doniach S 1986 Use of general potentials in multiple-scattering theory *Phys. Rev. A* **34** 4682–94